Graphs and Best Fit

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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1. A graph is an argument

A graph should make a relationship visible: proportionality, a threshold, a power law, an outlier or a systematic trend. Choose axes that match the physical question. Label units, state the scale, and show error bars when possible. A graph without units is a picture, not a measurement.

Linear graphs are especially useful because slope and intercept have clear meanings. If the physics predicts $y = mx + b$, then m and b can be compared with theory.

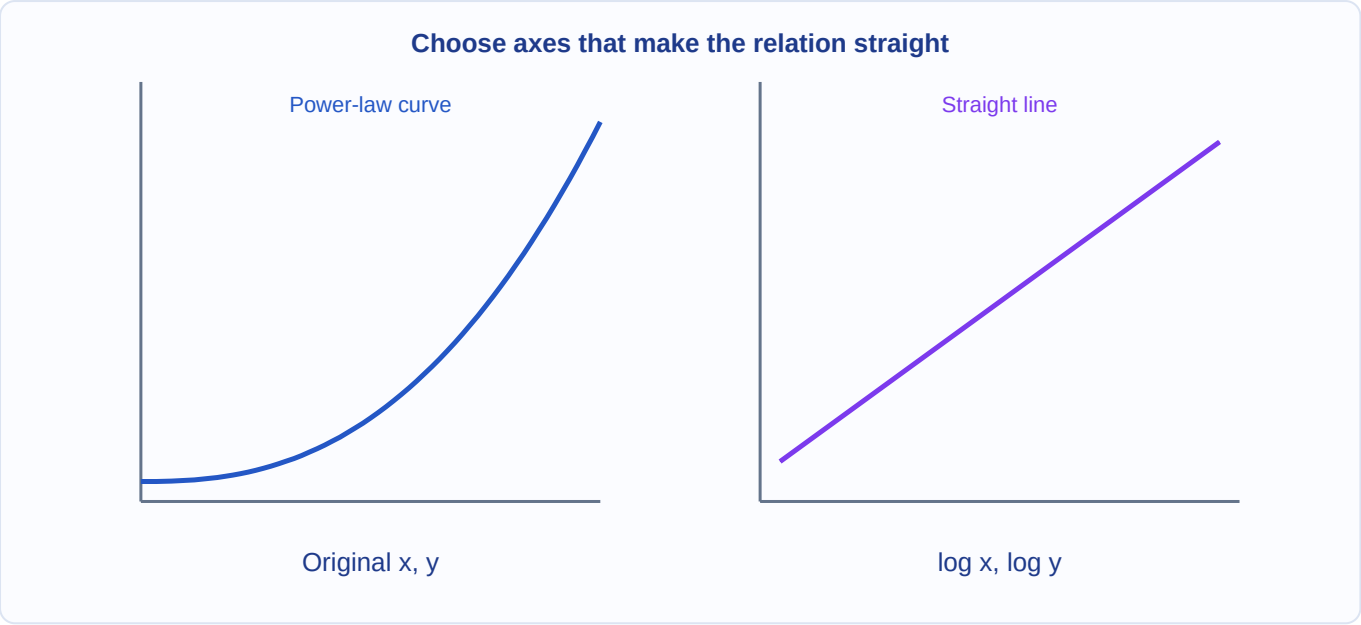


Figure 1. The slope and intercept of the transformed plot have physical meaning.

2. Linearization

Many laws are not linear in their original variables but become linear after a transformation. For $y = ax^n$ with $x, y > 0$, plot $\log y$ against $\log x$; the slope is n. For exponential decay $N = N_0 e^{-\lambda t}$ with $N > 0$, plot $\ln N$ against t; the slope is $-\lambda$. For a small-amplitude simple pendulum, $T^2 = (4\pi^2/g)L$, so plot T^2 against L.

The transformation must be applied to uncertainties too. A log plot compresses large values and expands small ones; error bars become asymmetric in the original variable.

3. Least-squares fitting

For a straight line $y = mx + b$, the least-squares slope and intercept are

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad b = \bar{y} - m\bar{x}. \tag{1}$$

This minimizes the sum of squared vertical residuals. Ordinary unweighted least squares assumes the x values are exact or much more precise than y, independent y errors with equal variance, and a correctly specified linear mean. If variances differ, use weights; if both axes have uncertainty, a more general fit may be needed.

The coefficient of determination is

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}. \tag{2}$$

A high R^2 does not prove the model is correct. It only says the line explains much of the variance.

4. Residuals and outliers

Plot residuals $y_i - \hat{y}_i$ against x or $\hat{y}$. Random residuals support the model. Curvature suggests a missing term. A funnel shape suggests changing variance. A single extreme residual may be an outlier, but investigate before deleting it.

Outliers can dominate a least-squares fit because residuals are squared. Robust fitting or a stated clipping rule can help, but transparency is essential.

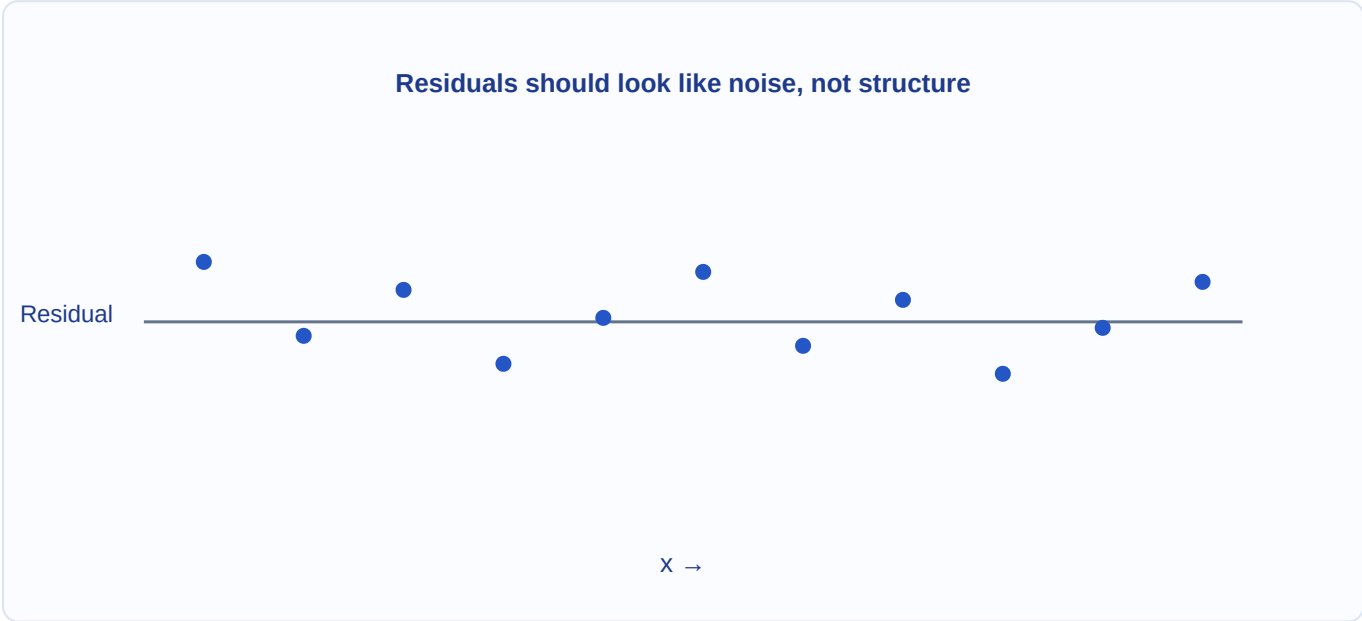


Figure 2. A high R^2 does not guarantee a good model.

5. Uncertainty in slope and intercept

For a simple unweighted fit with residual standard deviation $s = \sqrt{\sum (y_i - \hat{y}_i)^2/(n-2)}$ (two fitted parameters),

$$\sigma_m = \frac{s}{\sqrt{\sum (x_i - \bar{x})^2}}, \qquad \sigma_b = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2}}. \tag{3}$$

A wider x range reduces slope uncertainty. More points help, but a few well-placed points at the extremes can be more valuable than many clustered points.

6. From fit to physical conclusion

After fitting, check the intercept against physical expectation. A nonzero intercept may reveal a zero error. Compare the slope with theory including uncertainty. State the fitted model, the uncertainty, the residual behaviour and any limitations.

INTERACTIVE EXPERIMENT

Fit a line and inspect the residuals.

Synthetic data follow y = slope·x + 1 with Gaussian-like noise. The least-squares line, slope uncertainty and R^2 are computed live; a residual panel is drawn below the fit.

Fitted slope	Fitted intercept	R^2
1.999	**0.964**	**0.9991**

Slope uncertainty
0.019



Least squares minimizes squared vertical residuals. R^2 alone can hide outliers or curvature; always inspect residuals.

Try this. Turn on the outlier. How much do the slope and R^2 move? Why should you plot residuals, not only R^2?

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Plot the data first. Choose axes that linearize the physics. Fit only after the model is plausible. Inspect residuals. Report slope, intercept, uncertainty and limitations.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Slope from two points. A line passes through (1, 3) and (4, 9). Find the slope and intercept.

Worked solution →

[1] Problem 2 — Linearization. A pendulum obeys $T^2 = (4\pi^2/g)L$. What should be plotted to obtain a straight line, and what does the slope represent?

Worked solution →

[2] Problem 3 — Power law. Data follow $y = ax^n$. On a log-log plot the slope is 2.5 and the intercept is $\log_{10} a = 0.3$. Find n and a.

Worked solution →

[2] Problem 4 — Least-squares slope. For three points (0,1), (1,3), (2,5), compute the least-squares slope and intercept. Why is the fit exact?

Worked solution →

[2] Problem 5 — Residual. A fit gives $\hat{y} = 2x + 1$. For the point (3, 8), find the residual. Is the point above or below the line?

Worked solution →

[1] Problem 6 — R² caution. A data set has $R^2 = 0.99$ but the residuals show a clear curve. Is the linear model adequate? What should you do?

Worked solution →

[2] Problem 7 — Slope uncertainty. A fit has residual standard deviation $s = 0.2$ and $\sum(x_i - \bar{x})^2 = 10$. Estimate the slope uncertainty.

Worked solution →

[2] Problem 8 — Outlier effect. A single point is far from the line. Why can it strongly change a least-squares fit? What is a responsible way to handle it?

Worked solution →

[3] Problem 9 — Exponential decay. A decay curve gives a straight line on a $\ln N$ versus t plot with slope $-0.1\ \text{yr}^{-1}$. Find the decay constant and half-life.

Worked solution →

[3] Problem 10 — Physical intercept. A fit of measured length versus true length gives intercept 0.2 cm. What might this indicate? How would you test it?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.